

OPSBRAIN AI

ET AI HACKATHON 2.0

A grounded operational memory for industrial teams.

PS 8: AI for Industrial Knowledge Intelligence: Unified Asset & Operations Brain

Submission brief | Working prototype | July 2026

OpsBrain AI turns fragmented engineering records into connected evidence, then delivers maintenance, safety, compliance, and RCA answers with citations, confidence, and an inspectable retrieval trail.

Executive summary

Industrial teams keep critical knowledge in manuals, SOPs, incident reports, inspection notes, spreadsheets, and shift handovers. Finding evidence during a safety decision is slow, and generic chatbots cannot prove where an answer came from. OpsBrain automatically ingests operational documents, extracts metadata, creates adaptive chunks, builds entity relationships, and combines keyword, semantic, and graph retrieval. The Copilot answers from retrieved evidence and exposes the source document, page or chunk, confidence, related sources, and retrieval diagnostics.

The problem and the users

Users: maintenance engineers, reliability teams, EHS and safety officers, plant managers, compliance teams, and incident investigators. **Pain:** fragmented records, repeated failures, slow RCA, and poor auditability.

Product workflow

Step	Outcome
01 - Upload	PDF, DOCX, PPTX, XLSX, CSV, HTML, TXT, Markdown, or ZIP
02 - Understand	Extract content, metadata, sections, entities, and relationships
03 - Index	Create adaptive chunks for keyword, semantic, and graph retrieval
04 - Answer	Generate a grounded response with citations and confidence
05 - Act	Follow evidence through Assets, Graph, Compliance, RCA, and Insights

What the prototype demonstrates

Capability	Demonstrated implementation
Document intelligence	Multi-format extraction, metadata, adaptive chunks, and ingestion status
Enterprise RAG	Hybrid keyword/vector/graph retrieval, reranking, Gemini generation, and citations
Knowledge graph	Entity and relationship index for assets, incidents, procedures, departments, and regulations
Operational Copilot	Conversation context, source ranking, confidence, related documents, and retrieval trace
Asset, compliance, RCA	Connected asset history, evidence-led compliance gaps, and root-cause workflows
Product quality	Responsive desktop/mobile UI, authenticated routes, health endpoints, CI, and Docker topology

Architecture and innovation

The product is evidence-first: ingestion, retrieval, generation, and UI are connected through a FastAPI contract rather than hardcoded demo responses.

Layer	Role
React + Firebase	Responsive authenticated product experience
FastAPI gateway	REST APIs, health/readiness, ingestion, intelligence, and RAG routes
Document pipeline	Extraction, metadata, adaptive chunking, and incremental indexing
Hybrid retrieval	Keyword + semantic/vector + entity/graph evidence with reranking
Gemini orchestration	Dynamic grounded prompt, answer, citations, confidence, and fallback
Operations views	Copilot, Documents, Assets, Graph, Compliance, RCA, and Insights

Innovation highlights

Evidence-first Copilot: citations and confidence are first-class response fields. **Digital asset memory:** one asset view connects manuals, incidents, maintenance, risks, and recommendations. **Graph + RAG:** relationships retrieve context that plain keyword search misses. **Inspectable AI:** retrieval counts and source ranking let teams verify the answer path.

Demo script

1. Sign in with Firebase Email/Password. **2.** Open Documents. **3.** Ask: "What are the vibration alert thresholds for PMP-101?" **4.** Point out the cited document, chunk, confidence, and retrieval counts. **5.** Open the PMP-101 asset and Graph views. **6.** Finish with Compliance, RCA, and Insights.

Impact, verification, and submission

Expected impact

Reduce time spent searching records, improve consistency and auditability of maintenance and compliance decisions, reuse incident lessons across assets, and provide a practical foundation for QMS, sensor, and plant-system integrations.

Verification evidence

Check	Result
Backend tests	30 passed
Frontend quality	TypeScript validation and production Vite build passed
Runtime	Health endpoint and live Gemini-grounded RAG query verified
Delivery	GitHub repository, CI, Docker Compose, architecture, requirements matrix, deck source, and demo storyboard included

Submission checklist

Upload this PDF as the detailed document. Add the architecture diagram and presentation deck as additional files. Use the repository URL below. Record the demo using the script on page 3 and upload the 3–4 minute MP4 or a tested public link.

GitHub: <https://github.com/Pawan8010/ET-GEN-AI-HACKATHON>

Responsible scope

The working prototype demonstrates multi-format document intelligence, hybrid RAG, graph retrieval, citations, responsive UI, asset intelligence, compliance/RCA workflows, and live Gemini integration. Scanned-image OCR/P&ID; computer vision and external QMS connectors are documented extension points requiring plant drawings, model selection, or customer credentials. OpsBrain is decision support, not an autonomous safety controller.

Built for the ET AI Hackathon 2.0 | PS8